



Drivers' gaze patterns when resuming control with a head-up-display: Effects of automation level and time budget

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ABSTRACT

The removal of drivers' active engagement in driving tasks can lead to erratic gaze patterns in SAE Level 2 (L2) and Level 3 (L3) automation, which has been linked to their subsequent degraded take-over performance. To further address how changes in gaze patterns evolve during the take-over phase, and whether they are influenced by the take-over urgency and the location of the human-machine interface, this driving simulator study used a head-up display (HUD) to relay information about the automation status and conducted take-over driving experiments where the ego car was about to exit the highway with variations in the automation level (L2, L3) and time budget (2 s, 6 s). In L2 automation, drivers were required to monitor the environment, while in L3, they were engaged with a visual non-driving related task. Manual driving was also embodied in the experiments as the baseline. Results showed that, compared to manual driving, drivers in L2 automation focused more on the HUD and Far Road (roadway beyond 2 s time headway ahead), and less on the Near Road (roadway within 2 s time headway ahead); while in L3, drivers' attention was predominantly allocated on the non-driving related task. After receiving take-over requests (TORs), there was a gradual diversion of attention from the Far Road to the Near Road in L2 take-overs. This trend changed nearly in proportion to the time within the time budget and it exaggerated given a shorter time budget of 2 s. While in L3, drivers' gaze distribution was similar in the early stage of take-overs for both time budget conditions (2 s vs. 6 s), where they prioritized their early glances to Near Road with a gradual increase in attention towards Far Road. The HUD used in the present study showed the potential to maintain drivers' attention around the road center during automation and to encourage drivers to glance the road earlier after TORs by reducing glances to the instrument cluster, which might be of significance to take-over safety. These findings were discussed based on an extended conceptual gaze control model, which advances our understanding of gaze patterns around control transitions and their underlying gaze control causations. Implications can be contributed to the design of autonomous vehicles to facilitate the transition of control by guiding drivers' attention appropriately according to drivers' attentional state and the take-over urgency.

1. Introduction

The development of driving automation has forecasted a wide range of benefits for the transportation network, including higher road efficiency, improved driving comfort, and fewer accidents caused by human mistakes. Characterized by varied capabilities of performing dynamic driving tasks, 6 levels of driving automation are defined, ranging from no driving automation (Level 0, L0) to full driving automation (Level 5,

L5; see [SAE International, 2018](#) for detailed definitions). Nevertheless, there is a long way to go before full driving automation becomes a reality ([Shladover, 2015](#)), and we should only expect to see Level 2 and Level 3 (L2, L3; [SAE International, 2018](#)) systems in the transportation network in the short to medium term, with some L2 features already available in highways for the merit of low traffic complexity. At L2 (partial driving automation), drivers are relieved of the longitudinal and lateral control, yet not supervision, of the dynamic driving task, whereas

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L3 systems (conditional driving automation) further allow drivers to look away from the driving environment. However, in both cases, drivers are still responsible to take over control from automated driving, should the systems reach their limits. This drawback in L2 and L3 systems raises a major safety challenge that the removal of drivers' involvement in the perceptual-motor control loop (Mole et al., 2019) during automation may impair their perception and comprehension of the environment (Goncalves et al., 2020), which might ultimately compromise their take-over driving performance (Eriksson & Stanton, 2017; Gold et al., 2016; Li et al., 2021). As drivers heavily rely on visual information to support motor control of driving, knowing drivers' gaze strategies in the context of automation, especially during the transition of control, enhances our ability to tackle this challenge.

Existing empirical studies have revealed that automation affects gaze behaviours (Feldhütter et al., 2017; Louw et al., 2017; Louw & Merat, 2017; Mackenzie & Harris, 2015; Schnebelen et al., 2019). Here, reduced visual attention towards the road center is seen with an increase in automation level (Carsten et al., 2012; Louw & Merat, 2017), which is sometimes exacerbated by increased engagement with visual non-driving related tasks (Huisinigh et al., 2015; Louw et al., 2019; Zeeb et al., 2016). Drivers also tend to look further ahead in the road scene (Mackenzie & Harris, 2015) and scan more horizontally (Damböck et al., 2013; Louw & Merat, 2017) when they were not in active physical control of the vehicle. Such erratic gaze patterns and inattention to the road scene, diminishing drivers' situation awareness (Endsley and Kaber, 1999), have been demonstrated to cause more incorrect take-over responses with a higher collision rate (Louw et al., 2017; Zeeb et al., 2015). However, Feldhütter et al. (2017) revealed that a long period of automation and engagement with non-driving related tasks led to less cumulative gaze duration towards the road scene and longer reaction time, but no significant changes in terms of accelerations and time to collision. These controversial findings in existing studies regarding the link between changes in drivers' gaze patterns and the take-over performance might be attributed to Feldhütter et al. (2017)'s use of a notable longer time budget (6 s) as compared to Louw et al. (2017) (3 s) and Zeeb et al. (2015) (2.5–3.5 s). However, there is a lack in literature to address this postulation by exploring changes in drivers' gaze patterns during the transition period, under varying time budgets.

Louw et al. (2019) also investigated how drivers divided their attention across different areas of interest (Aols) not only during the automation but also during the transition of control, with or without an additional non-driving related task (visual arrow task). It adds new insights to literature by showing that the human-machine interface (HMI, integrated into the instrument cluster) displaying the status of the system occupied quite a proportion of attention during automation, and drivers further prioritized the HMI over the road scene for perception in the early stage of control transitions, regardless of whether they were performing the additional non-driving related task. This implies that more considerations should be devoted to ensuring that drivers' intuitive visual sampling from the HMI would not interfere with the detection of on-road incidents. A potential approach to diminish the issue of impaired situation awareness and the visual competence between the road scene and the HMI, in this context, might be head-up displays (HUDs) (Damböck et al., 2012; Feierle et al., 2019; Schömig et al., 2018). Conventional HUDs in vehicles project speedometer, tachometer, and other driving-related information to the windshield or somewhere ahead in the field of view, while more informative prototypes further overlay animations onto the surroundings in the form of augmented reality (Feierle et al., 2019). While HUDs have been proven to improve driving performance and cause less mental stress in manual driving (Liu, 2003; Liu & Wen, 2004), they also show merits of, in the context of automated driving, keeping drivers from looking away from the road scene when searching for information that would otherwise be presented on the instrument cluster (Schömig et al., 2018).

However, solely retaining drivers' attention on the road might not be sufficient, as gaze should also be well coordinated with motor control to

ensure safe driving (Mole et al., 2019). Research from manual driving (Kountouriotis et al., 2012; Land & Lee, 1994; Lappi et al., 2013; Lehtonen et al., 2014) has shown that drivers tend to execute overt stimulus sampling in a "move-dwell-move" manner by executing "guiding fixations" (GFs; on-road fixations allocated around 1–2 s time headway ahead of the ego car), and intermittently scanning even further ahead with interleaving "look-ahead fixations" (LAFs; fixations previewing considerably further ahead on the road than where GFs are directed). Literature generally suggests that GFs help determine impending motor commands for positional control in a short timescale, while LAFs serve for future hazard anticipation and trajectory planning (Lappi & Mole, 2018; Lehtonen et al., 2013, 2014; Mars et al., 2012; Mole et al., 2019; Wilkie et al., 2008). Such gaze strategy of alternating between GFs and LAFs, referred to as gaze polling behaviour, was also addressed in the circumstance of automated driving (Mars et al., 2012; Navarro et al., 2016; Schnebelen et al., 2019), which showed that drivers tend to executed fewer GFs and more LAFs (compared with manual driving). It was also found in Navarro et al., 2016 that even if drivers had kept their attention on the road and noticed the obstacle at similar times (so had a similar time budget to evade), they performed evasive maneuvers more aggressively in take-over driving, as compared to manual driving. Mole et al. (2019) attributed these observations to the decoupled gaze mode from steering caused by automation and inferred that there should be a recoupling period in the early stage of control resumptions, regardless of whether drivers were looking at the road upon receiving take-over requests. Through characterizing the dynamics of drivers' gaze polling behaviours, one can shed more light on the gaze recoupling process of how drivers altered their visual strategies dynamically to support physical control, which might be of extra significance to road safety under varying takeover urgency (manipulated by the length of the time budget). However, few attempts in literature have addressed this.

1.1. Gaze control theory

To better understand the underlying causes of the changes in drivers' gaze strategies, it is necessary to borrow some knowledge from the field of human gaze control. Existing literature generally interprets human's overt shift of attention from two dichotomous aspects: (1) the bottom-up orientation driven by the properties of stimuli (e.g., saliency) or (2) the top-down orientation controlled according to the prior knowledge or task requirements (Henderson, 2003; Shiferaw et al., 2019). A growing body of literature agrees on a theoretical integration of gaze control as a prediction process that reduces the error based on the external (bottom-up) information and internal (top-down) expectation, through selective visual sampling (Adams et al., 2013; Rauss & Pourtois, 2013; Shiferaw et al., 2019). In this context, to selectively sample information from the redundant visual world, the brain prioritizes the encoding of less frequent, but more salient, features by suppressing frequently occurring ones (Koch & Poggio, 1999; Shiferaw et al., 2019). In the absence of top-down modulation, locations with salient features are more likely to be fixated through the bottom-up influence. Meanwhile, the brain makes top-down predictions about the relevant incoming perceptual information to mediate oculomotor actions along with the bottom-up influence. These predictions originate from observers' internal state, e.g., intentions, and prior knowledge of the scene and task requirements. As observers' internal state can also be influenced by external information via the ascending bottom-up process, which, in turn, will also influence the bottom-up process through top-down modulation, the bottom-up and top-down processes are not mutually exclusive but closely interdependent (Shiferaw et al., 2019). These underpinnings of human gaze control have been incorporated into a conceptual gaze control model formulated by Shiferaw et al. (2019), which provides a more theoretical perspective to interpret drivers' gaze patterns in the circumstance of automated driving and control transitions. However, to the best of the authors' knowledge, no empirical studies on driving gaze behaviours have attempted to explain observations with these gaze control

underpins.

1.2. Current study

To fill the aforementioned research gaps, the current study aimed to provide a deeper understanding of changes in drivers' gaze patterns throughout the entire time domain around the transition of control (including both phases before and after and TORs), with variations in the automation level (L2 and L3) and time budget (2 s and 6 s). A HUD, positioned above the road path, was used to relay information about the automation system and indicate TORs, which aimed to encourage drivers to maintain their attention on the road. We expanded an existing conceptual framework of gaze control to reflect the circumstance of take-over driving, based on which theoretical explanations for the observed gaze patterns were formulated. In this paper, it was hypothesized that the effects of the automation level on drivers' gaze patterns during automation would carry over to the subsequent take-over phase, which should also vary with the length of time budget.

The rest of this paper is organized as follows: Section 2 presents the details of experiment conduction and data collection. Section 3 describes the procedure of data processing as well as the statistical methods used in this study. Section 4 illustrates experimental results, and this is followed by Section 5, where the effects of the automation level, time budget, and HUD on drivers' gaze patterns were discussed based on the conceptual gaze control model. Finally, key conclusions, implications, and limitations of this study are summarized in Section 6.

2. Methods

2.1. Apparatus

The experiments were conducted using the motion-based driving simulator of the State Key Laboratory of Automotive Safety and Energy at Tsinghua University, China (see Fig. 1). A full-scale BMW series-3 mock cab with all driving control operations is mounted on a 6-degree-of-freedom motion base. The projection system comprises 5 screens, providing a frontal field of view of approximately $200^\circ \times 50^\circ$ (horizontal $\times$ vertical) and a rearward field of view of $50^\circ \times 30^\circ$ from driver's perspective. All manual operational inputs, vehicle's dynamic status data, and its spatial positions in the simulated scenarios were recorded at a frequency of 60 Hz. SensoMotoric Instruments Eye Tracking Glasses 2 wireless (SMI ETG) were used to collect drivers' eye

movements, including raw gaze position, pupil size, and gaze vectors, at a sampling rate of 60 Hz, along with the frontal-view stimuli recorded at 30 Hz and 960×710 pixels. Physiological data, including electrodermal activity (EDA), electromyogram (EMG) and photoplethysmographic signals (PPG) were collected by the BIOPAC MP150 system at 1000 Hz. Subjective workload assessment data were also collected using the NASA Task Load Index (NASA-TLX, Hart & Staveland, 1988) questionnaire.

2.2. Participants

Following approval by the Tsinghua University Ethics Committee, 37 participants were recruited via the internet and advertising posters in Beijing, China. 11 participants were excluded from the analysis due to either simulator motion sickness (3 participants) or unreliable data collection (8 participants, participants closed their eyes for a while or/and adjusted the eye-tracking glasses during driving yielding uncalibrated gaze coordinates). The remaining 26 participants (6 females and 20 males) included 11 young (23–27 years old), 13 middle-aged (28–50 years old), and 2 old (>50 years old) drivers. Each held a valid driving license for at least 2 years (Mean = 7.04, SD = 5.30). The proportions for the short (5,000–10,000 km), medium (10,000–100,000 km), and long (>100,000 km) driving mileage were 34.6 % (n = 9), 46.2 % (n = 12), and 19.2 % (n = 5), respectively. Drivers were compensated 120 RMB for taking part.

2.3. Human-machine interface and take-over request

A head-up display was used to provide information about the automation status (see Fig. 2). During automation, a green steering wheel symbol superimposed by a WI-FI icon was shown in the central-upper region in the frontal field of view. A text message was also included beneath the symbol indicating that the L2/L3 automation was activated (in Chinese). A visual-auditory TOR was employed in this study. When it came to situations that required drivers to respond, the HMI changed accordingly (see Fig. 2d-e). Additionally, the interface flashed at a rate of 1 Hz and was accompanied by high-frequency auditory warning beeps.

2.4. Non-driving related task

To create L3 driving conditions, participants were asked to perform a visual non-driving related task: the Surrogate Reference Task (SuRT,

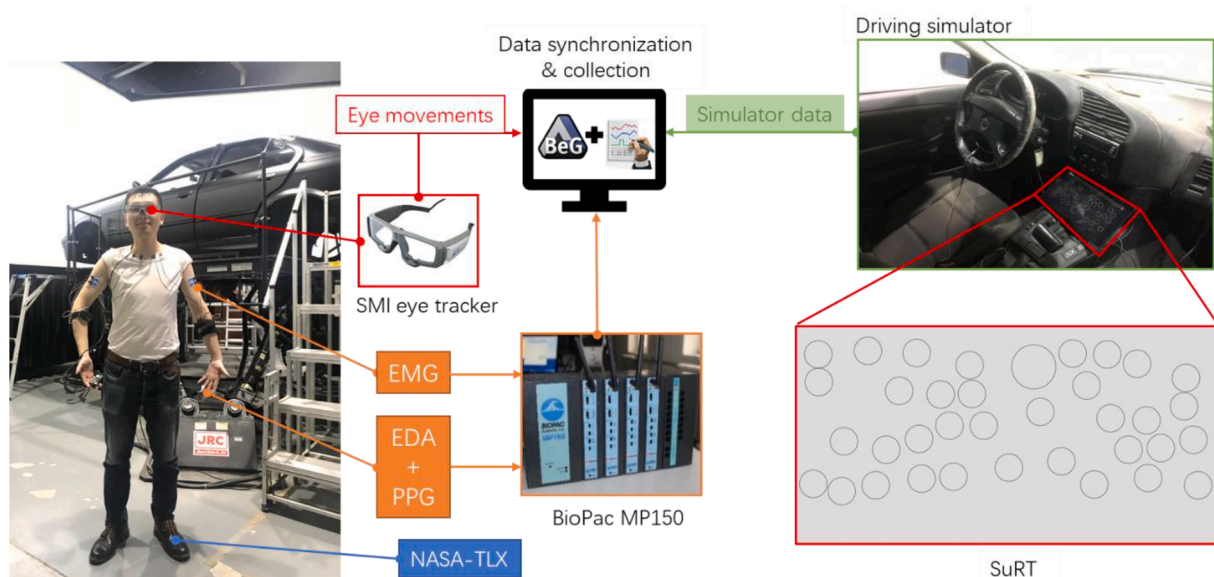


Fig. 1. The schematic for apparatus setup (SuRT: the interface of the Surrogate Reference Task, ISO/TS14198, 2012).

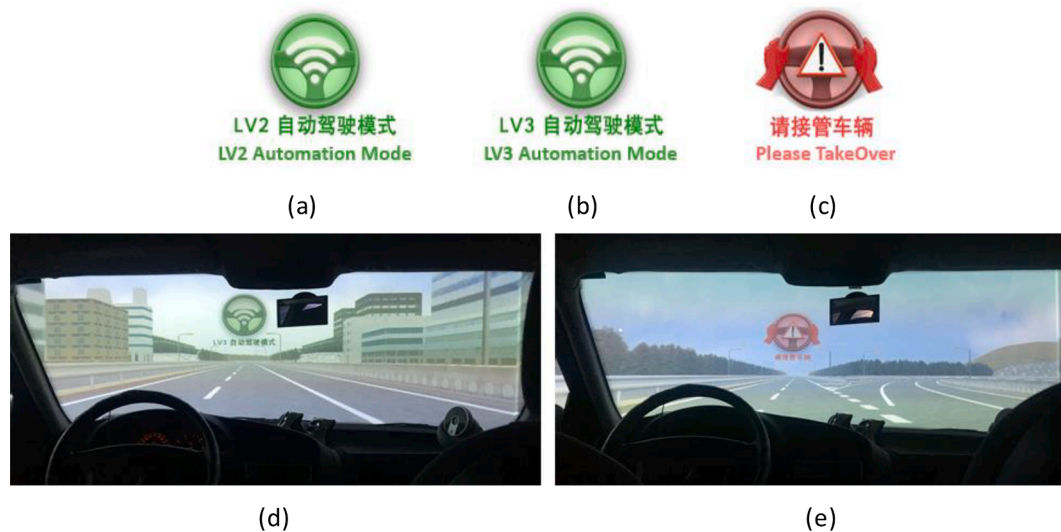


Fig. 2. The HMI designs. (a) L2 (Level 2 automation mode, in Chinese), (b) L3 (Level 3 automation mode, in Chinese), (c) TOR, (Please take over, in Chinese), (d-e) the HMI viewing from driver's perspective.

ISO/TS14198, 2012; see Fig. 1). This task required participants to identify a large target circle among other smaller ones by fixating on it. A research assistant who sat in the co-pilot seat performed the selection, based on drivers' gaze position, which was monitored in real-time via a subnotebook connected to the eye-tracking glasses. This adjustment was aimed to avoid potential artefacts to physiological signals caused by hand movements. The SuRT was implemented on a tablet placed at the bottom of the center console of the mock cap, such that drivers could not monitor the road scene via peripheral view when performing tasks.

2.5. Operating the automated driving system

In this study, the automation function was implemented using an MPC-based vehicle path-tracking algorithm (Li et al., 2017). The automation system was activated automatically when the vehicle entered its operational design domain (SAE International, 2018), keeping the vehicle driving in the center lane at 90 km/h. Upon activation, a voice message was broadcast (in Chinese) indicating that the automation was engaged. At the same time, the HUD displaying the L2/L3 automation symbol was shown to drivers. At this stage, participants were no longer responsible for the dynamic driving tasks and were allowed to move their feet and hands off the pedals and steering wheel, respectively. For L2, participants were required to monitor the environment during automation. While for L3, after around 20-s activation of the automation, a voice message announced the beginning of the SuRT (in Chinese). Participants then performed the SuRT and were not required to monitor the road.

Take-over events in this study occurred when the ego car was about to exit the operational design domain of the system, i.e., exiting the highway and entering a straight or right-turn off-ramp, as these events account for most of the situations in practice (Eriksson & Stanton, 2017; SAE International, 2018). When exiting the operational design domain, the automation system would disengage, and a visual-auditory TOR was issued. Participants were required to resume the control of the vehicle, both laterally and longitudinally, by steering, braking, or accelerating. The visual-auditory TOR was dismissed once participants gave any effective manual operational inputs, i.e., a steering input >5 degrees, a brake pressure reaching 0.5 MPa, or the ego car's acceleration exceeding 0.001 m/s². To avoid unexpected driver-initiated take-over operations, participants could not override the automation system unless a TOR was given.

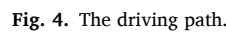
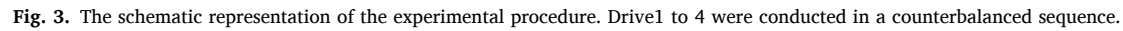
2.6. Experimental and scenario design

We used a $2 \times 2 \times 2$ within-participant design, with three independent variables: time budget (TB; 2 s, 6 s), automation level (L2, L3), and roadway curvature (straight, curved). As shown in Fig. 3, each participant completed a training-manual drive and four automated drives, which lasted about 100 min, altogether. The training-manual drive contained two sessions, i.e., the training session and the manual session. During the training session, participants practiced how to control the vehicle, perform the SuRT, and resume control from automation. This was followed by the manual session, during which drivers were asked to drive the car manually and data were recorded as the baseline for comparison with L2 and L3 conditions. As shown in Fig. 4, drives for L2 or L3 condition comprised four take-over trials, two for straight take-overs and two for curved ones, which occurred in an interleaving sequence. After each trial, the simulation was paused, followed by a NASA-TLX questionnaire inquiring subjective workload of the take-over trial just experienced. Then, the next trial followed. The driving scenarios for these five drives were similar, but the starting and ending points as well as the surrounding layouts of the road varied, to reduce participants' familiarity with the scenarios.

The highway comprised two normal driving lanes (3.5 m in width), and one emergency lane (3 m in width), in both directions. To minimize drivers' familiarity with the scenarios, the road included a mixture of curved and straight segments. Randomized off-road objects, including buildings, bridges, trees and mountains, were also incorporated. The speed range for manual driving, including the take-over periods, was 80–120 km/h, though drivers were encouraged to drive at around 90 km/h if they felt safe to do so. All TORs were prompted during the straight-approaching section before entering the off-ramps with a designed time budget (2 s, 6 s) (see Fig. 5). Participants were required to turn right or not, indicated by the layout of the traffic cones placed in the blocked direction. As road traffic can lead to highly road-event-dependent gaze patterns (Goncalves et al., 2020), which might interfere with the influences of the primary factors (automation level and time budget) of the current study, no ambient traffic was included in the driving environment.

2.7. Procedure

Upon arrival, participants were told the purpose of the experiment was to investigate their driving behaviours, eye movements, biological responses, as well as subjective assessments when driving in highway



The built-in SMI algorithm of the eye-tracking glasses can

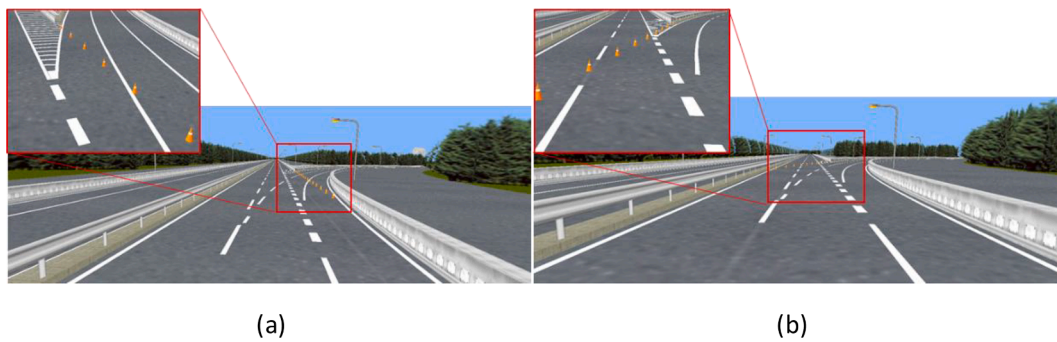


Fig. 5. The take-over scenarios. (a) Enter the straight ramp (the right-turn off-ramp was blocked), (b) Enter the right-turn off-ramp (the straight ramp was blocked).

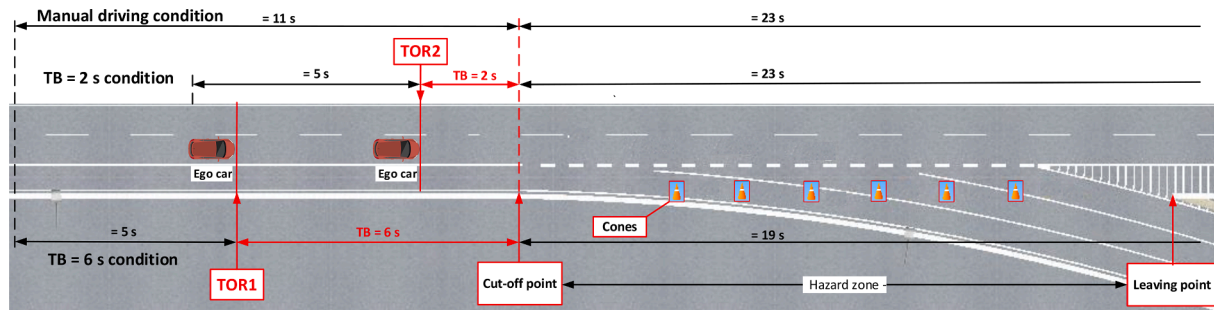


Fig. 6. The schematic of data segments for analysis. TOR1 represents the onset of the TOR given a TB of 6 s, and TOR2 represents the onset of the TOR given a TB of 2 s. The Cut-off point represents the geometric tangent point between the curved and straight ramps. The Leaving point represents the moment when the ego car left the off-ramp zone completely.

automatically categorize eye-movement events into fixations, saccades and blinks, and it can further merge adjacent pseudo fixations into a single one, in the presence of head motions (see [Instruments \(SMI\), 2017](#) for detailed illustrations). Raw gaze points belonging to fixations were remapped onto a static image depicting the frontal field of view, as well as a part of the interior of the cab, as shown in [Fig. 7](#). This gaze remapping process was implemented via the Semantic Gaze Mapping tool of BeGaze (the data analysis software that comes with the SMI eye-tracking glasses), which aimed to eliminate the influence of head motions so that gaze points were aligned into an identical coordinate system for all participants.

To understand how drivers distributed their attention among different regions, 13 areas of interest (AoIs) were defined, based on their real-world meaning (see [Fig. 7](#)). The Speedometer covered the area of the instrument cluster of the cab, the SuRT covered the area of the tablet running the task, the Center Console encompassed the center console excluding the SuRT area, the HUD encompassed the area for displaying both the symbol and text message, and the Left/Right Off Road encompassed the roadside surroundings. To investigate drivers' overt

gaze polling behaviours of shifting fixations between near and far roadway regions, the roadway was further divided into two AoIs, i.e., the Near Road and Far Road, where the Far Road denoted the roadway starting from around 50 m ahead of the car to the end of the road, as well as the spatial areas beneath the HUD area. Such that the dividing boundary between the Near Road and Far Road was inherently around 2 s in time headway when the driving speed was set at 90 km/h, which aimed to be in line with the existing observations that GFs are usually directed 1–2 s time headway ahead in routine driving ([Land & Lee, 1994; Lappi et al., 2013; Lehtonen et al., 2013](#)). Other regions included the Sky, Left/Right Mirrors, and Left/Right Window.

Time-series representation of gaze distribution, the distribution of drivers' first fixation location after TORs, the proportions of gaze time towards AoIs, gaze time and frequency of GFs and LAFs, and gaze polling frequency were used to assess how drivers allocate their attention during different phases around transitions.

The proportions of gaze time towards different AoIs (during the specific time windows of interest) were statistically analyzed using IBM SPSS Statistics 25. Kolmogorov-Smirnov and Shapiro-Wilk tests showed that not all measurements were normally distributed, even after transformations. Thus, the nonparametric Friedman test and Wilcoxon signed-rank test were used in statistical analysis. The Friedman test is a statistical test used to determine whether 3 or more measurements from the same group of participants significantly differ from each other. It is a non-parametric alternative to the commonly used one-way ANOVA (analysis of variance) with repeated measures when the assumption of the normal distribution is violated. Similarly, the Wilcoxon signed-rank test is the nonparametric test equivalent to the paired-sample *t*-test that was used to compare two sets of measurements from the same participants, while it also does not require measurements to meet the normal distribution assumption. In this study, Friedman tests were conducted separately for each time budget condition, to determine the simple main effects of the automation level (as the within-participant factor). Post-hoc pairwise comparisons were conducted using Wilcoxon signed-rank

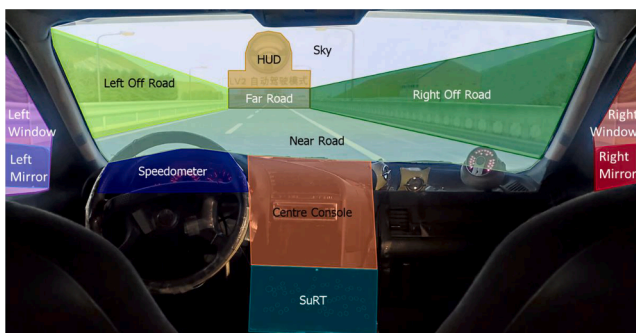


Fig. 7. The semantic segmentation of AoIs.

tests when the simple main effects of the automation level were significant. The simple main effects of time budget were also examined using Wilcoxon signed-rank tests. Bonferroni adjustments were applied to p -values in multiple pairwise comparisons. An α -value of 0.05 was adopted as the criterion for significance in all tests.

4. Results

Fig. 8 depicts the proportions of glances to each AoI, over time, across all driving conditions. Distinctions can be observed between different driving conditions. Generally, the gaze distribution during the manual drive was relatively stable, except for the period around the Cut-off Point, which indicated that the road event (the off-ramp and cones) altered drivers' gaze patterns. While for L2 and L3 drives, the gaze distribution evolved diversely with time (after TORs), which was also affected by the length of time budget. More specific presentations are shown in three sections: (1) Before TORs, (2) After TORs, and (3) Gaze polling behaviours.

4.1. Before TORs

To statistically compare the distribution of gaze to each AoI, for different automation levels, the proportions of gaze time to each AoI were calculated for the 5-s period before TORs (calculated as the total time looking to a particular AoI by the participant, divided by the length of the time window). Friedman tests showed that there was a significant effect of automation level ($p < 0.001$) on the proportions of gaze time over five main AoIs: Near Road, Far Road, HUD, Speedometer, and SuRT

(see Fig. 9 and Table 1).

In manual driving, the Far Road and Near Road areas received the largest and second-largest proportions of drivers' attention, with occasional glances towards the Speedometer. The average proportion of gaze time on these three AoIs occupied, respectively, 44.2 % (Far Road), 40.0 % (Near Road), and 4.0 % (Speedometer) of the total duration of this period. Compared to manual driving, drivers in L2 were more likely to gaze on the HUD (Mean = 11.2 %; $Z = -5.443$, $p < 0.001$), but less onto the Near Road (Mean = 10.5 %; $Z = -4.435$, $p < 0.001$). Considering the

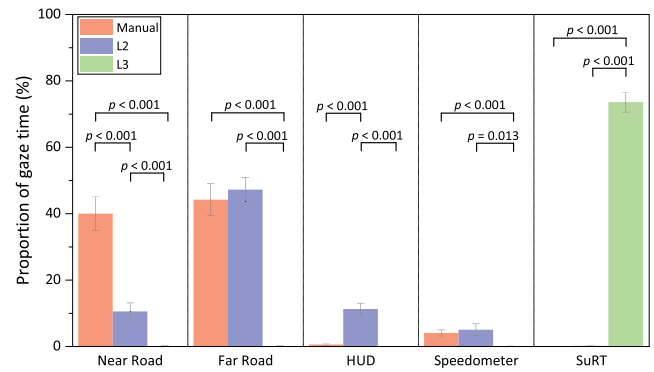


Fig. 9. The distribution of gaze time among different AoIs within the 5-s period before TORs (error bars stand for standard errors, p -values were adjusted with Bonferroni corrections in multiple pairwise comparisons).

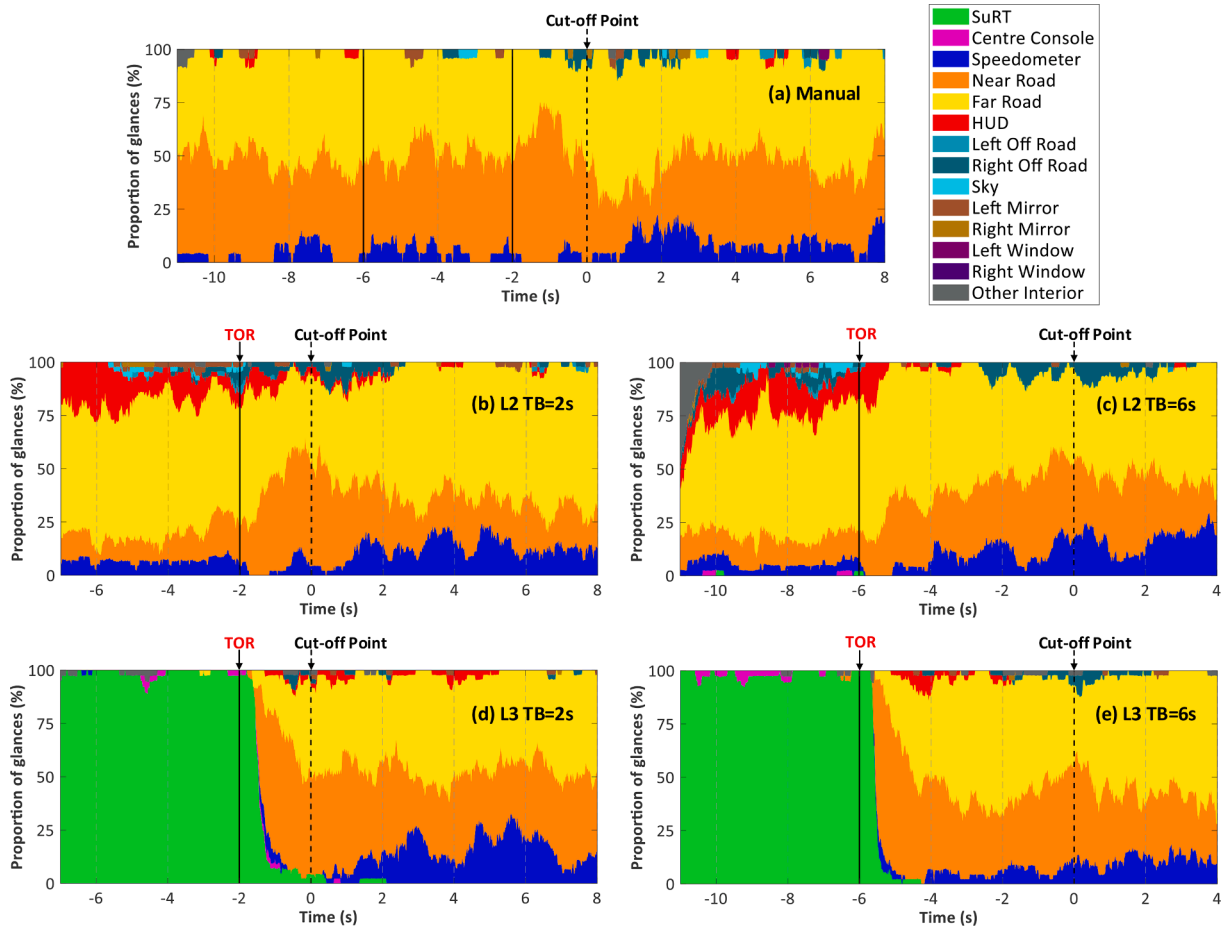


Fig. 8. The proportion of glances to each AoI, over time, across all driving conditions. Note: (1) although the HUD was invisible in manual driving periods, it represents the same area as those in L2 & L3 drives from drivers' perspective. (2) the solid black line and black dash line represent the onset of the TOR and the moment when the vehicle reached the Cut-off point of ramps (also illustrated in Fig. 6), respectively.

Table 1

Comparisons of the proportions of gaze time to different AoIs (within the 5-s period before TORs).

AoI	Friedman Test		Z (p) of Wilcoxon Signed-Rank Test		
	$\chi^2(df)$	p	L2-M	L3-M	L3-L2
Near Road	58.400 (2)	<0.001*	-4.435 (<0.001*)	-5.511 (<0.001*)	-4.937 (<0.001*)
Far Road	62.887 (2)	<0.001*	-0.483 (1.000)	-5.511 (<0.001*)	-6.154 (<0.001*)
HUD	76.200 (2)	<0.001*	-5.443 (<0.001*)	-1.604 (0.326)	-5.443 (<0.001*)
Speedometer	19.263 (2)	<0.001*	-0.335 (1.000)	-3.823 (<0.001*)	-2.845 (0.013*)
SuRT	101.351 (2)	<0.001*	-1.000 (0.952)	-6.215 (<0.001*)	-6.215 (<0.001*)

* $p < 0.05$, p -values were adjusted with Bonferroni corrections in multiple pairwise comparisons.

spatial distribution of AoIs (see Fig. 7), Fig. 8 also showed that when decoupled from physical control of the vehicle, drivers in L2 automation dispersed more glances towards non-road areas (Speedometer, Right Off Road, Left Off Road, Left Mirror, Right Mirror, and Sky), along with more concentrated glances around the road center areas (mostly covering the Far Road and HUD areas). While in L3, drivers' full engagement with the SuRT led to a high percentage of gaze (73.6 %) to the SuRT area, which differed significantly from both manual ($Z = -6.125$, $p < 0.001$) and L2 automated ($Z = -6.215$, $p < 0.001$) driving.

4.2. After TORs

After TORs, drivers' attention in L2 take-overs was redirected from the HUD and Far Road to the Near Road within the time budget (see Fig. 8). While in L3, drivers prioritized their fixations to the Near Road with a gradual increase in fixations to the Far Road area. For manual driving, although no explicit TORs were given, drivers showed a diversion of attention from the Far Road to the Near Road area, when the ego car was approaching the off-ramp sections of the road.

To understand how drivers redistributed their visual attention in the early stages after TORs, the proportions of gaze time to AoIs were calculated as a mean percentage of the first 2 s after TORs (Fig. 10). Results showed that, given a time budget of 2 s, drivers in L2 paid more attention to the Near Road area in the first 2 s after TORs (from 10.5 % to 35.5 %), as compared to the preceding 5-s period before TORs, at a notable cost of attention towards the Far Road (from 47.2 % to 39.9 %) as well as a minor decline of attention towards the HUD (from 11.2 % to 5.1 %) and the Speedometer (from 4.9 % to 3.5 %). A similar trend was also observed in the 6-s time budget condition for the Near Road (from 10.5 % to 23.7 %), HUD (from 11.2 % to 5.9 %), and Speedometer (from 4.9 % to 2.4 %) areas. However, attention over the Far Road area increased from 47.2 % to 57.4 %, in this case.

As for the L3 take-over conditions, the gaze distribution in the first 2 s after TORs was similar for both time budget conditions. The Near Road was prioritized for visual perception (occupying 33.3 % for TB = 2 s, and 36.5 % for TB = 6 s), while less attention was provided to the Far Road area (16.0 % for TB = 2 s, and 18.9 % for TB = 6 s). It is worth noting that the SuRT still captured a substantial proportion of drivers' attention in this period (>15 %), which might be due to the delay of gaze response after TORs.

Friedman tests also indicated that there was a significant effect of the automation level on the proportions of gaze time to the Far Road, HUD, and SuRT, but not to the Near Road and Speedometer (see Table 2), in the first 2 s after TORs, for both time budget conditions. Post-hoc Wilcoxon signed-rank tests showed that drivers in L3 take-overs were less likely to fixate towards the Far Road area than in L2 take-overs and manual driving, while no significant difference was found between the latter two cases (see Table 2). In the case of manual driving, drivers

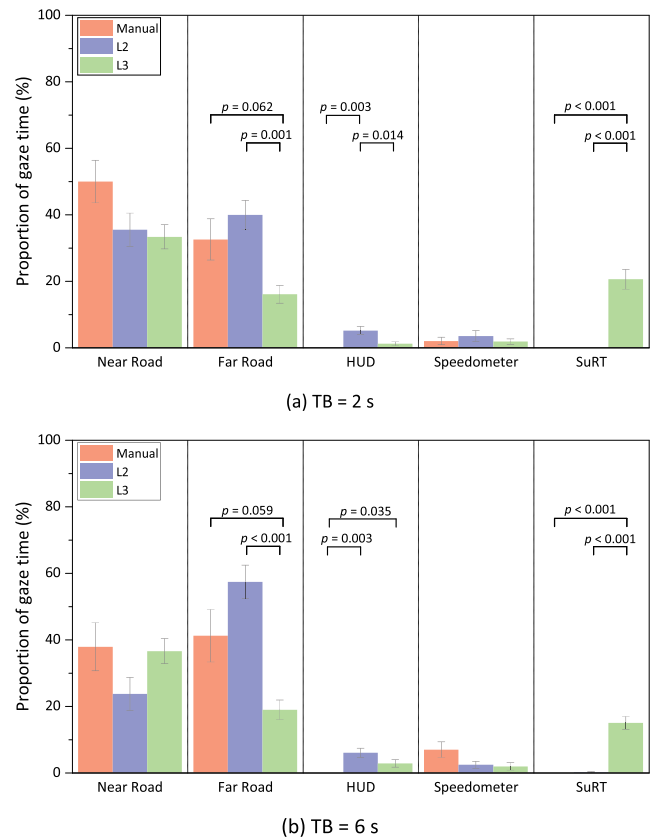


Fig. 10. The proportion of gaze time among different AoIs within the first 2-s period after TORs (error bars stand for standard errors, p -values were adjusted with Bonferroni corrections in multiple pairwise comparisons).

performed no fixations towards the HUD area (the HUD was not visible to drivers in this case). Drivers in L2 take-overs paid more attention to the HUD area than that in L3 ($Z = -2.840$, $p = 0.014$), when given a time budget of 2 s. In terms of the SuRT area, due to the delay of gaze response, more gaze time was observed in L3 take-overs than that in manual and L2 take-over driving (for both time budget conditions).

The effects of time budget on the proportions of gaze time among different AoIs within the first 2 s after TORs were shown in Table 3. It was found that, for L2 take-overs, when given a longer time budget of 6 s, drivers paid less attention to the Near Road ($Z = -2.322$, $p = 0.020$) and more to the Far Road ($Z = -3.010$, $p = 0.003$). However, for L3 takeovers, there was no significant difference between the two time budget conditions ($p > 0.05$ for each AoI).

In terms of the location of drivers' first fixation (see Fig. 11), similar conclusions could also be drawn that drivers in L3 take-overs fixated tremendously more towards the Near Road immediately after TORs, while in L2 their attention was more likely to remain on the Far Road and HUD areas.

4.3. Gaze polling behaviours

To understand how drivers' gaze polling behaviours were impacted by automated driving and how it evolved under different take-over urgency, fixations towards the Near Road and Far Road were hereinafter referred to as Guiding Fixations (GFs) and Looking-ahead Fixations (LAFs), respectively, and a valid gaze polling event was defined, following a similar approach as that used by Schnebelen et al. (2019) (see Fig. 12): (1) a saccade (an eye movement jumping from one fixation to another) was launched from the Near Road (GF area) going through any other AoIs and finally landed in the Far Road (LAF area), (2) attention remained in the Far Road area for one or several fixations, (3)

Table 2

The effects of automation level on the proportions of gaze time among different AoIs (within the first 2-s period after TORs).

TB	AoI	Friedman Test		Z (p) of Wilcoxon Signed-Rank Test		
		$\chi^2(df)$	p	L2-M	L3-M	L3-L2
2 s	Near Road	4.832(2)	0.089	/	/	/
	Far Road	9.260(2)	0.010*	−0.749 (1.000)	−2.312 (0.062)	−3.511 (0.001*)
	HUD	18.292(2)	<0.001*	−3.300 (0.003*)	−1.826 (0.204)	−2.840 (0.014*)
	Speedometer	4.541(2)	0.103	/	/	/
	SuRT	50.000(2)	<0.001*	0 (1.000)	−4.373 (<0.001*)	−4.373 (<0.001*)
6 s	Near Road	4.337(2)	0.114	/	/	/
	Far Road	12.641(2)	0.002*	−1.613 (0.320)	−2.327 (0.059)	−4.127 (<0.001*)
	HUD	16.536(2)	<0.001*	−3.297 (0.003*)	−2.521 (0.035)	−2.012 (0.133)
	Speedometer	3.733(2)	0.155	/	/	/
	SuRT	49.368(2)	<0.001*	−1.000 (0.952)	−4.373 (<0.001*)	−4.373 (<0.001*)

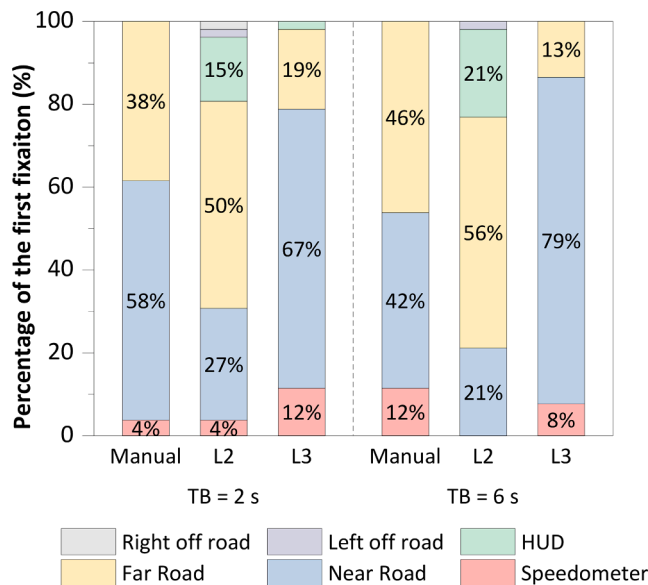
*p < 0.05, p-values were adjusted with Bonferroni corrections in multiple pairwise comparisons.

Table 3

The effects of time budget on the proportions of gaze time among different AoIs (within the first 2-s period after TORs).

AoI	Z (p) of Wilcoxon Signed-Rank Tests	
	L2	L3
Near Road	−2.322 (0.020*)	−0.559 (0.576)
Far Road	−3.010 (0.003*)	−0.821 (0.411)
HUD	−0.542 (0.588)	−1.481 (0.139)
Speedometer	−1.126 (0.260)	−0.491 (0.889)
SuRT	−1.000 (0.317)	−1.830 (0.067)

*p < 0.05.

**Fig. 11.** The distribution of drivers' first fixation location after TORs.

attention shifted back to the Near Road (GF area) via a single saccade or several saccades through any other AoIs. The proportions of gaze time during the sliding time window (calculated as the mean percentage of gaze time during a 2-s sliding time window, Fig. 13) and frequency of GFs and LAFs (number of fixations per second: N/s, calculated in a 2-s sliding time window over time, Fig. 14), and gaze polling frequency (number of gaze polling events per second: N/s, calculated in 2-s intervals over time, Fig. 15), were computed.

Results showed that, although drivers were required to keep their eyes on the road in L2 automation, they performed gaze polling less before TORs, as compared to manual driving (see Fig. 15), which was accompanied by more LAFs but fewer GFs (see the grey curve vs. red and green curves in Fig. 13a and Fig. 14a). Immediately after TORs, more

gaze polling events were detected in all take-over conditions, but in L2, take-overs with a time budget of 2 s yielded the highest frequency of gaze polling (0.48 N/s), which was quite distinct from the other conditions. For L2 take-over driving, an apparent trade-off between GFs and LAFs could be observed roughly within the time window (the interval between the TOR and the Leaving point). Specifically, more GFs and fewer LAFs (in terms of both the gaze time and fixation frequency, see Fig. 13a and Fig. 14a) were performed as the ego car approached the Cut-off point, which was then followed with fewer GFs and more LAFs. This trade-off between GFs and LAFs (roughly within the time budget) was scaled with the length of time budget (see L2 curves in Fig. 13a), while it evolved in a similar trend after the Cut-off point for both time budget conditions. Besides, the increase of GFs and decrease of LAFs (in terms of both gaze time and fixation frequency, see red curves in Fig. 13a and Fig. 14a) occurred about 2 s before the onset of the TOR given a time budget of 2 s, which indicated that drivers might have already anticipated the incoming take-over event in advance in this case. Moreover, after experiencing automated driving, drivers still tended to gaze more towards the LAF area and less towards the GF area, as compared to manual driving, even though they had resumed physical control for quite a while.

As for L3 take-over driving, drivers' gaze polling frequency, frequency and gaze time of GFs and LAFs were almost the same immediately after TORs (independent of the time budget, see L3 curves in Fig. 13b Fig. 14b and Fig. 15b). However, afterwards, these measures evolved quite distinctively in different time budget conditions. Given a longer time budget of 6 s, GFs shaped another apparent maximal roughly 1 s after TORs (point A in Fig. 13b and Fig. 14b) apart from the maximal near the Cut-off point, and it took around 3.5 s (point B in Fig. 13c and Fig. 14c) after TORs for GFs and LAFs to reach similar levels as those in L2 take-overs. Afterwards, GF/LAF curves of L2 and L3 take-overs evolved in similar trends (see Fig. 13c and Fig. 14c). However, given a shorter time budget of 2 s, there were much fewer GFs and more LAFs as compared to the 6-s time budget condition, throughout the period after TORs (see Fig. 13b and Fig. 14b).

5. Discussion

5.1. An integrated gaze control model during take-over driving

The current study investigated drivers' gaze behaviours across manual driving, L2, and L3 automated driving conditions, as well as the subsequent gaze patterns during the transition of control, under different time budgets. To have a better understanding of the observations, the conceptual gaze control model, as highlighted in Section 1.1, given by Shiferaw et al. (2019), was extended in this study by incorporating some components reflecting the circumstance of take-over driving (see Fig. 16 for schematic depiction, new components incorporated in this study were highlighted in yellow blocks). In this framework, gaze control is a prediction process, based on the interaction of bottom-

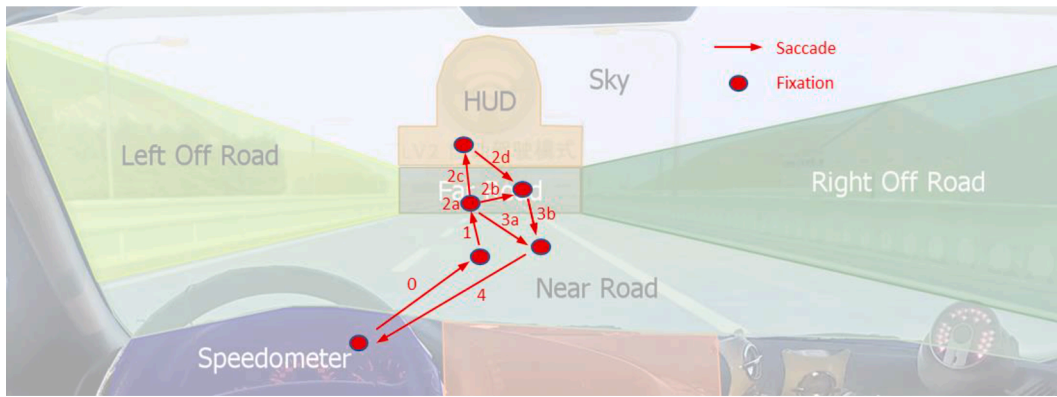


Fig. 12. Example of a schematic representation of gaze polling events (a valid event: 1 → 2 → 3).

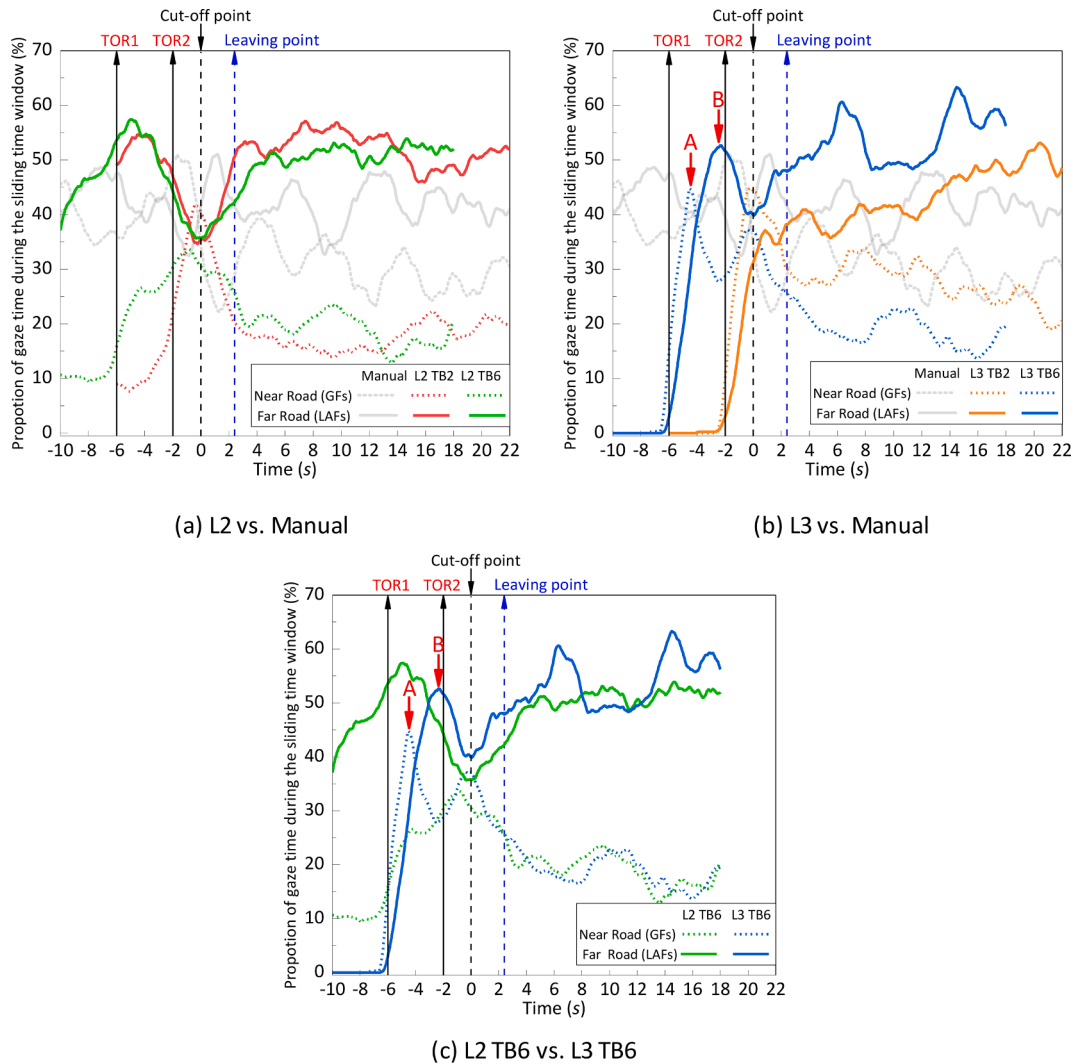


Fig. 13. The proportion of gaze time of GFs and LAFs. The Y-axis represents the mean proportion of gaze time during a 2-s sliding time window over time. TOR1 stands for the onset of the TOR given a TB of 6 s, TOR2 stands for the onset of the TOR given a TB of 2 s. The Cut-off point and Leaving point were illustrated in Fig. 6.

up and top-down flows of information. The salient features of stimuli that thrive the bottom-up influence could come from the HUD, or the off-ramps and traffic cones, due to their distinguishable colors, shapes, or dynamic patterns, such as twinkling or approaching. The internal state where the top-down prediction originates can be modulated by the information held in working memory, which ranges from the temporal

comprehension of the environment (referred to as situation awareness, Endsley and Kaber, 1999) to the prior knowledge of task requirements (Shiferaw et al., 2019). Situation awareness is built up based on the processing of external inputs including not only the sampled visual information but also information in other modalities (e.g., the auditory TOR, the dash red line in Fig. 16). In a take-over situation, drivers

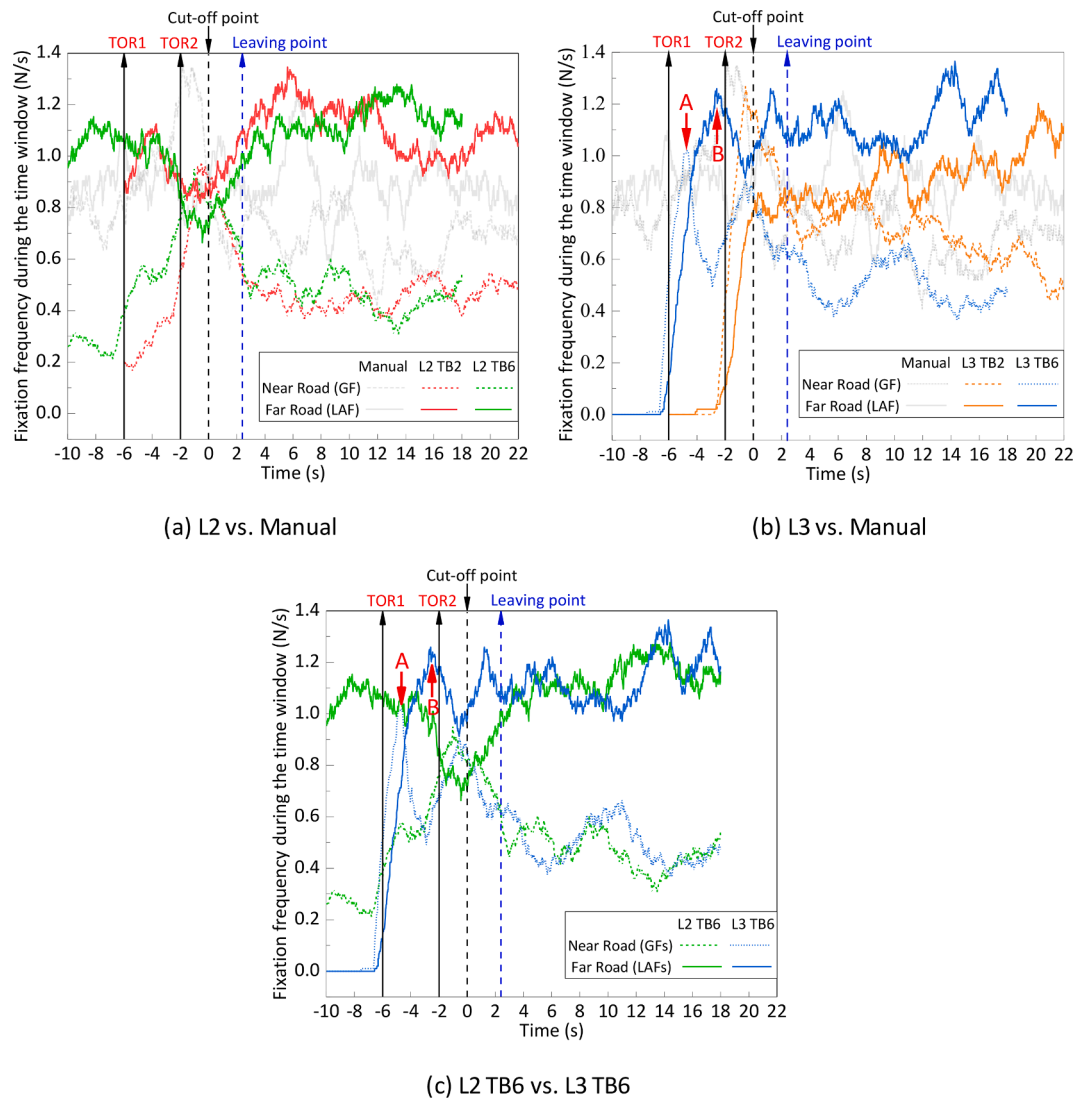


Fig. 14. Fixation frequency of GFs and LAFs. The Y-axis represents the mean fixation frequency in a 2-s sliding time window over time. TOR1 represents the onset of TORs given a TB of 6 s, TOR2 represents the onset of TORs given a TB of 2 s. The Cut-off point and Leaving point were illustrated in Fig. 6.

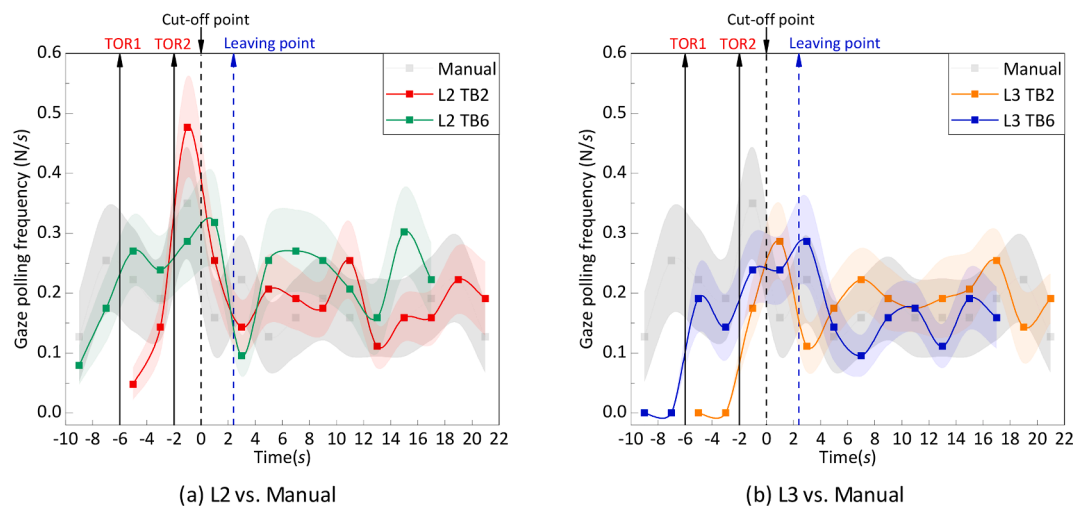


Fig. 15. Gaze polling frequency over time. The shaded areas represent standard errors. TOR1 represents the onset of TORs given a TB of 6 s, TOR2 represents the onset of TORs given a TB of 2 s. The Cut-off point and Leaving point were illustrated in Fig. 6.

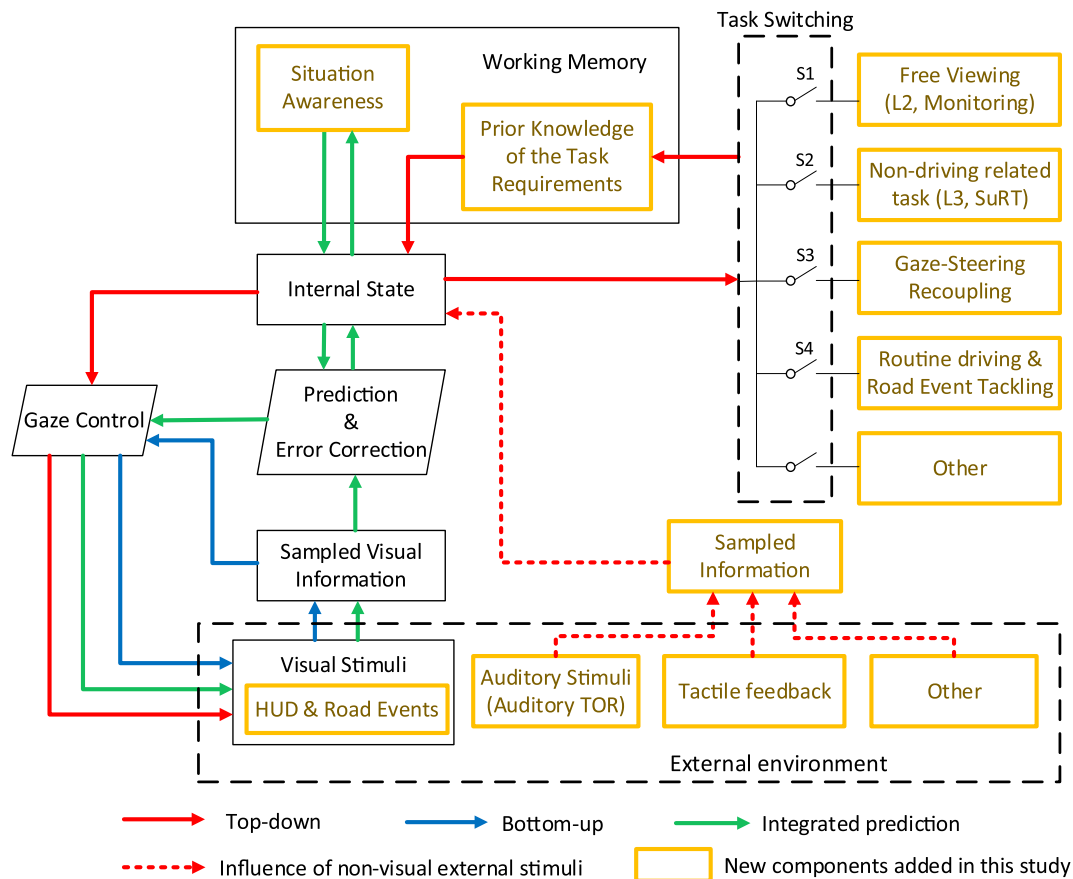


Fig. 16. The schematic framework of gaze control in take-over driving. The solid blue line reflects bottom-up influence; the solid red line refers to top-down influence; the dashed red line refers to the influence of the auditory TOR; the integrated prediction (solid green line) demonstrates that gaze control is a dual process integrating top-down and bottom-up interactions.

perform one or several tasks switching among environment monitoring (L2), non-driving related task performing (L3, SuRT), gaze-steering recoupling (in the early stage of control resumption), routine driving (lane-keeping in our experiments), and road event tackling. Thus, the top-down influence on gaze control will alter, in accordance with the task requirements and prior knowledge, which ultimately influences drivers' gaze behaviours. Key findings in our experiments will be discussed from three aspects based on this conceptual gaze control model: the relationship between (1) automation level and gaze patterns before TORs, (2) automation level and gaze patterns immediately after TORs, and (3) time budget and gaze polling patterns after TORs.

5.2. Automation level and gaze patterns before TORs

Findings showed that drivers' gaze patterns during automation differed significantly from manual driving. During L3 automation, drivers mainly focused on the SuRT, as they were told that there was no need for monitoring the environment in this case. A more important finding was that when relinquishing physical control to L2 automation, drivers redirected their gaze from the GF area (Near Road) to the LAF area (Far Road), Speedometer, HUD, as well as other areas beyond the road (compared with manual driving). The increase in attention towards non-road areas during L2 automation was in line with existing literature (Louw et al., 2019; Louw & Merat, 2017). Moreover, the diversion of attention from the GF area to the LAF area caused by vehicle automation in straight road driving has also been observed by studies concerning driving in a bend (Mackenzie & Harris, 2015; Schnebelen et al., 2019). However, in this study, much lower gaze polling frequency was observed in L2 automation compared with manual driving, which differed from

the observations in Schnebelen et al., 2019. This should be attributed to the variations in vehicle kinematic motions between driving on straight roads and curved roads, which, as Mackenzie & Harris (2015) claimed, affected the placement of drivers' vision.

The distinctions between L2 and manual driving in our experiments could be due to the difference in task requirements. Unlike manual driving, drivers' primary task in L2 automation switched from active lane-keeping and road event tackling to passive monitoring (switching from S4 to S1 in Fig. 16). Thus, the top-down modulation intended for active gaze-steering coordination was weakened, and the priority for GFs closely serving for ongoing positional control (Lappi & Mole, 2018) was inherently lowered. In the meantime, the top-down commands, reflecting the need for environmental monitoring, albeit mild, encouraged drivers to redistribute more gazes to the LAF area, for hazard detection/anticipation, as well as to the HUD and Speedometer, for system status checking. Concomitantly, the bottom-up influence was also enhanced, since the symbolic HUD concept used here was quite salient compared to the background of the environment, even in the peripheral view. In this way, the integrated effects of both top-down and bottom-up information in L2 not only released drivers' attention from the near roadway (GFs) but also guided their attention to the HUD when drivers were looking forward. These implications demonstrate that the layout of our HUD concept has the potential to maintain drivers' attention on the road for road-event detection, which is of extra significance to road safety in the case of take-overs in L2 driving.

5.3. Automation level and gaze patterns immediately after TORs

The preference for the AoIs to be fixated first when receiving TORs

should be dominated by a different gaze control mechanism, specifically, ascribed to the different levels of situation awareness in L2 and L3 take-overs. During L2 automation, as highlighted above, drivers made more glances to the central road area (mostly covering the LAF area and the HUD). Once a TOR was issued, the HUD altered from a constant green symbol to a twinkling red one, which increased the saliency of the HUD and would attract drivers' attention through the bottom-up influence of gaze control. In the meantime, suffering from the delay due to cognitive processing, new top-down cues reflecting the comprehension of the unexpected TOR (both visual and auditory) and changes of drivers' primary task (switching from monitoring to gaze-steering recoupling, then to routine driving) would thrive later than the bottom-up. Thus, the top-down influence remained to be originated based on previous temporal comprehension of the scene (situation awareness). On this basis, drivers' preference for the first fixation in L2 take-overs mainly reflected the changes in bottom-up inputs, while the changes in top-down influence emitted its influence in subsequent gaze behaviours. However, these implications should be underpinned by two assumptions (1) the incoming scenery information was not salient enough to compete with the TOR symbol and (2) drivers had not predicted the incoming TOR based on historical situation awareness. This can also be re-affirmed from Fig. 11 where the preference for the HUD and Far Road (LAF area) regarding the first fixation location was less prominent given a shorter time budget in L2 take-overs. Because in this case (L2, TB = 2 s), the off-ramp and traffic cones were more visible around the onset of TORs, hence, some drivers might have already anticipated the imminent TOR in advance (due to the learning effect after multiple trials), and they altered top-down commands accordingly.

Regarding drivers' first fixation distribution in L3, the dominant priority of attention towards the GF area was mainly driven by the top-down orientation. Once drivers heard the auditory TORs, they would have realized something happened but failed to predict the nature of the unfolding event (due to reduced situation awareness in L3) until they switched their attention back to the road. In this circumstance, a safer visual strategy was to dedicate early glances towards the road path in short time headway (GF area), as a potential hazard in this area would be more dangerous to the ego car and GFs could support ongoing positional control timely to minimize the danger. Thus, the top-down commands reflecting this gaze strategy based on drivers' prior knowledge/experience, would dominantly guide their first fixation towards the GF area.

In Louw et al., 2019, it was revealed that there was a visual competence between the road scene and the instrument-cluster-based HMI immediately after automation failure. Drivers in Louw et al., 2019 prioritized the HMI to the road scene for information perception, in the early stage of take-overs, which raised the concern that the perception from the HMI might impede drivers from looking back to the road timely. However, in the current study, the HUD concept partially undertook the function of the instrument-cluster-based HMI by relaying information of the automation system's status. Thus, the top-down influence reflecting this prior knowledge lowered the demand for checking with the Speedometer but increased the priority for road scene and HUD perception. Correspondingly, since the HUD could be easily observed when looking forward even using peripheral view, more attention to the roadway and much less to the Speedometer (the instrument cluster) were found as compared to observations in Louw et al., 2019, during the crucial early stage after TORs (the first 2-s period). This implication is of greater significance for L3 take-over driving where drivers had much-reduced situation awareness in the beginning.

5.4. Time budget and gaze polling patterns after TORs

The impact of time budget on drivers' gaze patterns was different for L2 and L3 take-overs. For L2 take-overs, the time budget mainly made a difference on gaze polling patterns in the period roughly between the TOR and the Cutting-point, where drivers gradually diverted their

glances from the LAF area to the GF area. While in L3, drivers prioritized attention towards the GF area, with a gradual increase in attention to the LAF area in the early stage of control transition, which was then followed by distinctive evolving trends for different time budget conditions. These different trends of the trade-off between GFs and LAFs indicated that the gaze recoupling process (Mole et al., 2019) was influenced by the interaction between the take-over urgency (manipulated by the time budget) and drivers' prior situation awareness (manipulated by the automation level).

As the ego car approached the off-ramps, three aspects contributed to the need for more GFs in both L2 and L3 take-overs: (1) the task requirement for gaze-steering recoupling in the early stage of take-overs, (2) the approaching off-ramps and traffic ones, and (3) higher pressure demanding for smooth lane-keeping before entering the hazard zone (the area between Cut-off point and Leaving point in Fig. 6). Taking the L2 take-over with 6 s of time budget for instance, initially, because drivers had decoupled gaze from steering during automation (fewer GFs and more LAFs as compared to manual driving), the top-down influence reflecting changes in task requirements (switching from S1 to S3 in Fig. 16) encouraged more GFs to guide ongoing positional control. Afterwards, as the off-ramp and traffic cones approached and became more visible and salient, an increasing proportion of glances could be attracted through the influence of top-down (driven by the need for road event tackling) and bottom-up (driven by the distinguishing saliency of the road event) interaction. This proportion of attention dedicated to the on-road event was mainly merged into the category of LAFs when the off-ramps and traffic cones still lay in the LAF area from driver's perspective and then regarded as GFs when the ego car continued to move ahead. Besides, the closer the off-ramp was, the higher pressure it applied to drivers for lane-keeping, which also prioritized the (top-down) demand for motor control and resulted in more GFs. These three processes took place sequentially when given a long time budget of 6 s, which caused the gradual attentional diversion from LAFs to GFs to be almost proportional with the lapse of time. Similarly, given a shorter time budget of 2 s, these three processes partially overlapped in time-scale, which made the time-proportional diversion from LAFs to GFs to be scaled with the length of time budget (L2 curves in Fig. 13a).

However, for the L3 situation, due to lack of situation awareness, the task requirement for gaze-steering recoupling in the early stage of take-overs required drivers to perform both more GFs (for ongoing positional control) and LAFs (for further hazard anticipating or detecting), whereby the situation awareness level also increased. Given the time budget of 6 s, this recoupling process could be distinguished from Fig. 13b and Fig. 14b (blue curves) where the GF curves reached the first maximal (point A in Fig. 13b and Fig. 14b) about 1 s after the TOR (for TOR1 in Fig. 13b and Fig. 14b), and then decreased until around 3.5 s after the TOR (for TOR1 in Fig. 13b and Fig. 14b) with the LAF curves continuously rising to the first maximal (point B in Fig. 13b and Fig. 14b). Afterwards, the attention diversion from LAFs to GFs took place and shaped in a similar trend as that in L2 take-overs (see the trend of curves after point B in Fig. 13c and Fig. 14c), which was also caused by the latter two aspects contributing to the increase of GFs (discussed in the immediately preceding paragraph). While given a shorter time budget of 2 s, both GFs and LAFs competed for attention in a limited time budget. Thus, drivers prioritized GFs ahead of LAFs when approaching the off-ramps and traffic cones, which led to much fewer LAFs than other take-over conditions even after the ego car had left the Leaving point for quite a while (see orange curves in Fig. 13b and Fig. 14b).

Thus, based on preceding discussions, it can be implied that a time budget of 6 s was long enough to enable drivers in L3 to gain as good situation awareness as that in L2 take-over driving in our scenarios. However, cautions must be applied when interpreting the time for situation awareness rebuilding with the specific value, as this might be highly dependent upon the complexity of the scenario and the length of time budget (Gold et al., 2013, 2016, 2017). Furthermore, when developing the automation systems, the minimum leading time for

error/failure detection, which inherently determines the minimal time budget, should not be simply determined based on the minimum time required for drivers to rebuild situation awareness. On the one hand, as the findings indicated, drivers' gaze behaviours were highly time-budget dependent, especially when the time pressure was high. On the other hand, in a more urgent circumstance where drivers must execute explicit maneuvers, such as obstacle avoiding or braking, additional time headway would be necessary to cushion the motion of the ego car.

6. Conclusion

This driving simulator study aimed to investigate drivers' gaze patterns around the transition of control, under varying automation levels and time urgency, where a HUD concept located above the roadway was introduced as an HMI to provide TOR messages. A conceptual gaze control model from literature (Shiferaw et al., 2019), based on the interaction of top-down and bottom-up influence, was extended to theoretically explain the observations of drivers' gaze patterns in take-over driving.

Findings confirmed the previous observations (Louw et al., 2019; Louw & Merat, 2017) that L2 automation elicited more voluntary glances beyond the road scene as the top-down modulation for active gaze-steering coordination was weakened when drivers' primary task changed to passive monitoring, and engaging with non-driving related tasks led to further reduced attention dedicated to the roadway. However, the driving risk caused by such decoupled gaze patterns might be mitigated by the proposed HUD concept in this study, as it showed the potential to maintain drivers' attention to the road center during automated driving and encourage drivers to glance the road earlier by reducing glances to the instrument cluster immediately after TORs. On straight roads, drivers executed similar gaze polling behaviours of negotiating between GFs and LAFs as those reported in curved-road sections (Mars et al., 2012; Schnebelen et al., 2019), and more LAFs were observed during L2 automation (Schnebelen et al., 2019).

Findings also confirmed the hypothesis that the effects of the automation level on drivers' gaze patterns during the automation phase carried over to the subsequent take-over phase, which also varied with the length of the time budget. Specifically, after TORs, drivers prioritized their attention, firstly towards different areas for different automation levels, i.e., largely remained on the HUD and LAF area (Far Road) in L2, while tremendously dedicated to the GF area (Near Road) in L3. This was driven by different gaze control mechanisms depending on drivers' prior situation awareness during automation. And this carry-over effect resulting from where drivers were looking before TORs also influenced drivers' subsequent gaze polling patterns in different time-scales, which was also modulated by the length of time budget. For L2 take-overs, the time budget for responses mainly made a difference on gaze polling patterns roughly within the period of the time budget, where drivers gradually diverted their glances from the LAF area to the GF area. And this attentional diversion changed nearly in proportion to the lapse of time. While in L3 situations, drivers' attentional distribution was similar in the early stage of take-overs for both time budget conditions (2 s vs. 6 s). They prioritized their early glances to the GF area with a gradual increase in attention towards the LAF area, which then was followed by distinctive evolving trends in different time budget conditions. In our experimental scenarios, a time budget of 6 s was long enough to enable drivers in L3 to regain as good situation awareness as that in L2 take-over driving. While given a shorter time budget of 2 s, both GFs and LAFs would compete for attention throughout the period within time budget in L3 take-overs, and drivers prioritized GFs ahead of LAFs at this stage, which might lead to insufficient hazard detection/anticipation from far roadway ahead.

By interpreting drivers' gaze patterns from the perspective of gaze control and gaze-steering coordination, this work advances our understanding of drivers' gaze strategies around automation transitions and the underlying gaze control causations. These findings provide

important implications for the design of autonomous vehicles, particularly in terms of how to ensure a safe and smooth take-over control by guiding drivers' attention appropriately according to their attentional state and the take-over urgency.

There are also some limitations in the present study. First, there was no take-over driving condition where the HUD was absent, which hindered the authors from examining the effect of the HUD directly from the performed experiments. However, by comparing our results with Louw et al., 2019's, the effects of the HUD were addressed, which were also explained theoretically based on the gaze control model. Second, the small sample size in the present study hindered the authors from further examining the effect of driver characteristics such as age and experience. Finally, this study has not considered the effects of traffic conditions on drivers' gaze patterns, such as traffic density and road geometry. However, these factors have been proven to have a significant influence on take-over performance (Gold et al., 2016; Louw et al., 2019). Further research is required to systematically address whether and how variations in driver characteristics and traffic conditions will interfere gaze patterns during take-over driving.

CRedit authorship contribution statement

Chengliang Xu: Conceptualization, Investigation, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing. **Tyron L. Louw:** Investigation, Methodology, Validation, Writing – review & editing. **Natasha Merat:** Investigation, Supervision, Writing – review & editing. **Penghui Li:** Conceptualization, Investigation, Writing – review & editing. **Mengxia Hu:** Validation, Investigation, Writing – review & editing. **Yibing Li:** Supervision, Project administration, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Adams, R.A., Shipp, S., Friston, K.J., 2013. Predictions not commands: active inference in the motor system. *Brain Struct. Funct.* 218 (3), 611–643. <https://doi.org/10.1007/s00429-012-0475-5>.
- Carsten, O., Lai, F.C.H., Barnard, Y., Jamson, A.H., Merat, N., 2012. Control task substitution in semiautomated driving: does it matter what aspects are automated? *Hum. Factors* 54 (5), 747–761. <https://doi.org/10.1177/0018720812460246>.
- Damböck, D., Weißgerber, T., Kienle, M., & Bengler, K. (2012). Evaluation of a contact analog head-up display for highly automated driving. *4th International Conference on Applied Human Factors and Ergonomics*. San Francisco, USA.
- Damböck, D., Weißgerber, T., Kienle, M., & Bengler, K. (2013). Requirements for cooperative vehicle guidance. *16th International IEEE Conference on Intelligent Transportation Systems (ITSC 2013)*, 1656–1661. 10.1109/ITSC.2013.6728467.
- Endsley, M.R., Kaber, D.B., 1999. Level of automation effects on performance, situation awareness and workload in a dynamic control task. *Ergonomics* 42 (3), 462–492.
- Eriksson, A., Stanton, N.A., 2017. Takeover time in highly automated vehicles: noncritical transitions to and from manual control. *Hum. Factors* 59 (4), 689–705.
- Feierle, A., Beller, D., Bengler, K., 2019. Head-up displays in urban partially automated driving: effects of using augmented reality. *IEEE Intel. Transp. Syst. Conf. (ITSC)* 2019, 1877–1882. <https://doi.org/10.1109/ITSC.2019.8917472>.

- Feldhütter, A., Gold, C., Schneider, S., Bengler, K., 2017. How the duration of automated driving influences take-over performance and gaze behavior. In: Schlick, C.M., Duckwitz, S., Flemisch, F., Frenz, M., Kuz, S., Mertens, A., Mütze-Niewöhner, S. (Eds.), *Advances in ergonomic design of systems, products and processes*. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 309–318.
- Gold, C., Damböck, D., Lorenz, L., Bengler, K., 2013. "Take over!" How long does it take to get the driver back into the loop? *Proc. Human Factors Ergon. Soc. Ann. Meet.* 57 (1), 1938–1942. <https://doi.org/10.1177/1541931213571433>.
- Gold, C., Körber, M., Lechner, D., Bengler, K., 2016. Taking over control from highly automated vehicles in complex traffic situations: the role of traffic density. *Hum. Factors* 58 (4), 642–652. <https://doi.org/10.1177/0018720816634226>.
- Gold, C., Happee, R., Bengler, K., 2017. Modeling take-over performance in level 3 conditionally automated vehicles. *Accid. Anal. Prev.* <https://doi.org/10.1016/j.aap.2017.11.009>.
- Goncalves, R.C., Louw, T.L., Quaresma, M., Madigan, R., Merat, N., 2020. The effect of motor control requirements on drivers' eye-gaze pattern during automated driving. *Accid. Anal. Prev.* 148, 105788 <https://doi.org/10.1016/j.aap.2020.105788>.
- Hart, S.G., Staveland, L.E., 1988. Development of NASA-TLX (task load index): results of empirical and theoretical research. In: Hancock, P.A., Meshkati, N. (Eds.), *Advances in Psychology*, Vol. 52. North-Holland, pp. 139–183. [https://doi.org/10.1016/S0166-4115\(08\)62386-9](https://doi.org/10.1016/S0166-4115(08)62386-9).
- Henderson, J.M., 2003. Human gaze control during real-world scene perception. *Trends Cogn. Sci.* 7 (11), 498–504. <https://doi.org/10.1016/j.tics.2003.09.006>.
- Huisingh, C., Griffin Jr, R., 2015. The prevalence of distraction among passenger vehicle drivers: a roadside observational approach. *Traffic Inj. Prev.* 16 (2), 140–146. <https://doi.org/10.1080/15389588.2014.916797>.
- Instruments (SMI), S. (2017). *BeGaze manual: Version 3.7*. Author Boston, MA.
- Koch, C., Poggio, T., 1999. Predicting the visual world: Silence is golden. *Nat. Neurosci.* 2 (1), 9–10. <https://doi.org/10.1038/4511>.
- Kountouriotis, G.K., Floyd, R.C., Gardner, P.H., Merat, N., Wilkie, R.M., 2012. The role of gaze and road edge information during high-speed locomotion. *J. Exp. Psychol. Hum. Percept. Perform.* 38 (3), 687–702. <https://doi.org/10.1037/a0026123>.
- Land, M.F., Lee, D.N., 1994. Where we look when we steer. *Nature* 369 (6483), 742–744. <https://doi.org/10.1038/369742a0>.
- Lappi, O., Lehtonen, E., Pekkanen, J., Itkonen, T., 2013. Beyond the tangent point: gaze targets in naturalistic driving. *J. Vis.* 13 (13), 11. <https://doi.org/10.1167/13.13.11>.
- Lappi, O., Mole, C., 2018. Visuomotor control, eye movements, and steering: a unified approach for incorporating feedback, feedforward, and internal models. *Psychol. Bull.* 144 (10), 981–1001. <https://doi.org/10.1037/bul0000150>.
- Lehtonen, E., Lappi, O., Kotkanen, H., Summala, H., 2013. Look-ahead fixations in curve driving. *Ergonomics* 56 (1), 34–44. <https://doi.org/10.1080/00140139.2012.739205>.
- Lehtonen, E., Lappi, O., Koirikivi, I., Summala, H., 2014. Effect of driving experience on anticipatory look-ahead fixations in real curve driving. *Accid. Anal. Prev.* 70, 195–208. <https://doi.org/10.1016/j.aap.2014.04.002>.
- Li, R., Li, Y., Li, S. E., Burdet, E., & Cheng, B. (2017). *Driver-automation indirect shared control of highly automated vehicles with intention-aware authority transition*. 26–32.
- Li, Q., Hou, L., Wang, Z., Wang, W., Zeng, C., Yuan, Q., Cheng, B., 2021. Drivers' visual-distraction take-over performance model and its application on adaptive adjustment of time budget. *Accid. Anal. Prev.* 154, 106099 <https://doi.org/10.1016/j.aap.2021.106099>.
- Liu, Y.-C., 2003. Effects of using head-up display in automobile context on attention demand and driving performance. *Displays* 24 (4), 157–165. <https://doi.org/10.1016/j.displa.2004.01.001>.
- Liu, Y.-C., Wen, M.-H., 2004. Comparison of head-up display (HUD) vs. head-down display (HDD): driving performance of commercial vehicle operators in Taiwan. *Int. J. Hum. Comput. Stud.* 61 (5), 679–697. <https://doi.org/10.1016/j.ijhcs.2004.06.002>.
- Louw, T., Madigan, R., Carsten, O., Merat, N., 2017. Were they in the loop during automated driving? Links between visual attention and crash potential. *Inj. Prev.* 23 (4), 281–286. <https://doi.org/10.1136/injuryprev-2016-042155>.
- Louw, T., Kuo, J., Romano, R., Radhakrishnan, V., Lenné, M.G., Merat, N., 2019. Engaging in NDRTs affects drivers' responses and glance patterns after silent automation failures. *Transport. Res. F: Traffic Psychol. Behav.* 62, 870–882. <https://doi.org/10.1016/j.trf.2019.03.020>.
- Louw, T., Merat, N., 2017. Are you in the loop? Using gaze dispersion to understand driver visual attention during vehicle automation. *Transp. Res. C: Emerg. Technol.* 76, 35–50.
- Mackenzie, A.K., Harris, J.M., 2015. Eye movements and hazard perception in active and passive driving. *Vis. Cogn.* 23 (6), 736–757. <https://doi.org/10.1080/13506285.2015.1079583>.
- Mars, F., Navarro, J., Paterson, K., 2012. Where we look when we drive with or without active steering wheel control. *PLoS One* 7 (8), e43858.
- Mole, C.D., Lappi, O., Giles, O., Markkula, G., Mars, F., Wilkie, R.M., 2019. Getting back into the loop: the perceptual-motor determinants of successful transitions out of automated driving. *Hum. Factors* 61 (7), 1037–1065. <https://doi.org/10.1177/0018720819829594>.
- Navarro, J., François, M., Mars, F., 2016. Obstacle avoidance under automated steering: Impact on driving and gaze behaviours. *Transport. Res. F: Traffic Psychol. Behav.* 43, 315–324. <https://doi.org/10.1016/j.trf.2016.09.007>.
- Rauss, K., Pourtois, G., 2013. What is bottom-up and what is top-down in predictive coding? *Front. Psychol.* 4 <https://doi.org/10.3389/fpsyg.2013.00276>.
- SAE International, 2018. Taxonomy and definitions for terms related to driving automation systems for on-road motor vehicles. *SAE Int.* https://doi.org/10.4271/J3016_201806.
- Schnebelen, D., Lappi, O., Mole, C., Pekkanen, J., Mars, F., 2019. Looking at the road when driving around bends: influence of vehicle automation and speed. *Front. Psychol.* 10 <https://doi.org/10.3389/fpsyg.2019.01699>.
- Schömig, N., Wiedemann, K., Naujoks, F., Neukum, A., Leuchtenberg, B., & Vöhringer-Kuhnt, T. (2018). An Augmented Reality Display for Conditionally Automated Driving. *Adjunct Proceedings of the 10th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 137–141. 10.1145/3239092.3265956.
- Shiferaw, B., Downey, L., Crewther, D., 2019. A review of gaze entropy as a measure of visual scanning efficiency. *Neurosci. Biobehav. Rev.* 96, 353–366. <https://doi.org/10.1016/j.neubiorev.2018.12.007>.
- Shladover, S. E. (2015). Road vehicle automation history, opportunities and challenges. *Mini-Seminar 'developments Selfdriving Vehicles in USA'*, 9.
- TS14198, 2012. Road vehicles-Ergonomic aspects of transport information and control systems-Calibration tasks for methods which access driver demand due to the use of invehicle systems. ISO TC22/SC13/WG8.
- Wilkie, R.M., Wann, J.P., Allison, R.S., 2008. Active gaze, visual look-ahead, and locomotor control. *J. Exp. Psychol. Hum. Percept. Perform.* 34 (5), 1150–1164. <https://doi.org/10.1037/0096-1523.34.5.1150>.
- Xu, C., Li, P., Li, Y., Merat, N., Lu, Z., & Guo, X. (2022). *Takeover Performance and Workload under Varying Automation Levels, Time Budget and Road Curvature*. 5th International Conference on Mechanics, Simulation and Control (ICMSC 2022), Xi'an.
- Zeeb, K., Buchner, A., Schrauf, M., 2015. What determines the take-over time? An integrated model approach of driver take-over after automated driving. *Accid. Anal. Prev.* 78, 212–221. <https://doi.org/10.1016/j.aap.2015.02.023>.
- Zeeb, K., Buchner, A., Schrauf, M., 2016. Is take-over time all that matters? The impact of visual-cognitive load on driver take-over quality after conditionally automated driving. *Accid. Anal. Prev.* 92, 230–239. <https://doi.org/10.1016/j.aap.2016.04.002>.